

EXPERIMENT 7

Build an Artificial Neural Network (ANN) using Keras/TensorFlow

1.Dataset Source

The dataset used in this experiment is the **Iris Flower Dataset**.

Source:

UCI Machine Learning Repository

<https://archive.ics.uci.edu/ml/datasets/iris>

It is also available through Scikit-learn's built-in dataset module.

2.Dataset Description

The Iris dataset is a multi-class classification dataset used for pattern recognition and supervised learning.

Dataset Characteristics:

- Total Samples: 150
- Number of Features: 4
- Number of Classes: 3
- Type: Structured numerical dataset

Features:

1. Sepal Length (cm)
2. Sepal Width (cm)
3. Petal Length (cm)
4. Petal Width (cm)

Target Variable:

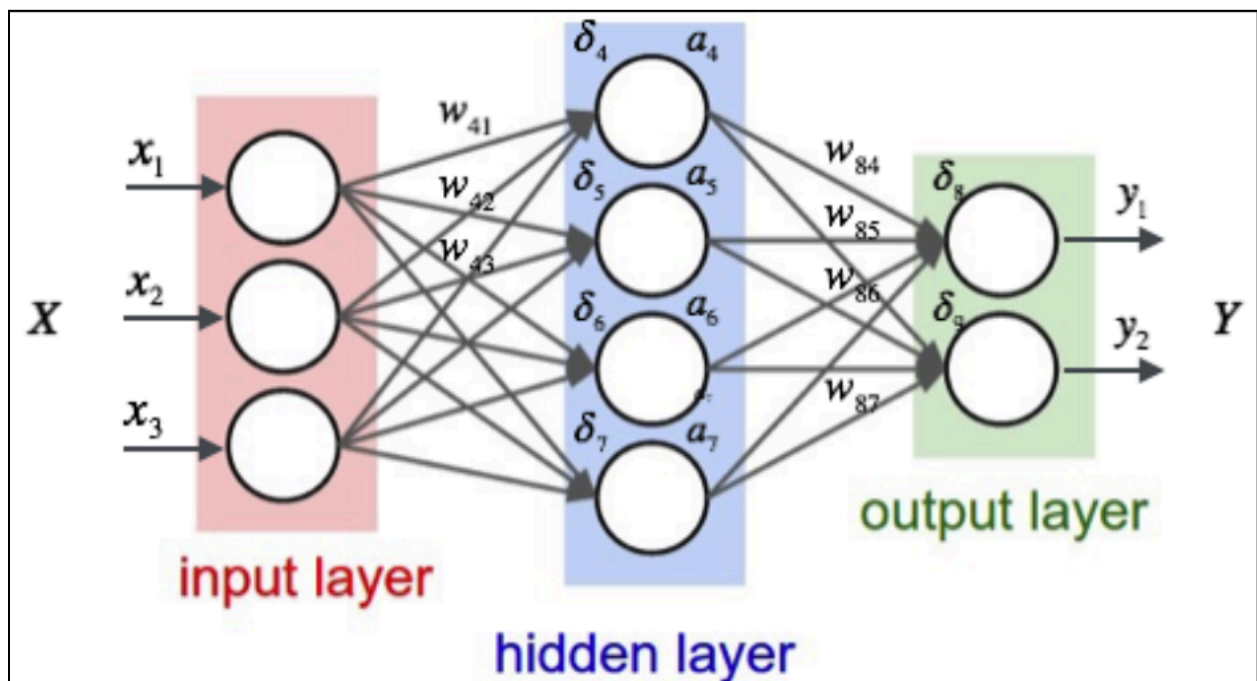
Flower Species:

- Setosa
- Versicolor
- Virginica

The target variable is categorical and converted into one-hot encoded format for ANN training.

The dataset is balanced, meaning each class has 50 samples

Ex :

**3. Mathematical Formulation of ANN**

An Artificial Neural Network consists of neurons arranged in layers.

For a single neuron:

$$Z = W.X + b$$

Where:

- W = weights
- X = input features
- b = bias

Activation function:

$$A = f(Z)$$

In this experiment:

Hidden Layer Activation:

ReLU

$$f(x) = \max(0, x)$$

Output Layer Activation:

Softmax

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum e^{z_j}}$$

Loss Function:

Categorical Crossentropy

$$Loss = - \sum y \log(\hat{y})$$

Optimizer:

Adam (Adaptive Moment Estimation)

The objective of ANN is to minimize the loss using backpropagation and gradient descent.

4. Algorithm Limitations

Although ANN is powerful, it has several limitations:

1. Requires large amounts of data for better performance.
2. Computationally expensive compared to simple models.
3. May suffer from overfitting.
4. Requires proper hyperparameter tuning.
5. Hard to interpret (Black-box model).

ANN may not perform well when:

- Dataset is very small
- Data is highly noisy
- Features are not scaled properly

5. Methodology / Workflow

The following steps were performed in this experiment:

Step 1: Data Loading

Iris dataset was loaded using Scikit-learn.

Step 2: Data Preprocessing

- One-hot encoding of target variable
- Train-test split (80% training, 20% testing)
- Feature scaling using StandardScaler

Step 3: Model Building

- Input Layer (4 neurons)
- Hidden Layer 1 (16 neurons, ReLU)
- Hidden Layer 2 (12 neurons, ReLU)
- Output Layer (3 neurons, Softmax)

Step 4: Model Compilation

- Optimizer: Adam
- Loss: Categorical Crossentropy
- Metric: Accuracy

Step 5: Model Training

- Epochs: 100
- Batch Size: 8
- Validation Split: 20%

Step 6: Model Evaluation

- Test accuracy calculated
- Accuracy and loss graphs plotted

Workflow Diagram :

Data → Preprocessing → ANN Model → Training → Evaluation → Result

6. Performance Analysis

The model performance was evaluated using:

Accuracy

Training Accuracy $\approx$ 97–99%

Validation Accuracy $\approx$ 95–96%

Test Accuracy $\approx$ 95–100%

This indicates that the model successfully learned patterns from the dataset.

Loss Analysis

- Training loss decreased gradually.
- Validation loss remained stable.
- No significant overfitting observed.

Interpretation

High test accuracy indicates that ANN effectively classified the three flower species. Since Iris dataset is well-structured and linearly separable, ANN performs exceptionally well.

7. Hyperparameter Tuning

Several hyperparameters were adjusted to improve performance:

1. Number of Hidden Layers

Experimented with 1 and 2 hidden layers.
Two hidden layers produced better accuracy.

2. Number of Neurons

Tested 8, 16, and 32 neurons.
16 neurons in first layer and 12 in second gave stable results.

3. Learning Rate (Adam optimizer)

Default learning rate performed well.
A very high learning rate caused unstable training.

4. Epochs

Tested 50, 100, and 150 epochs.
100 epochs provided good convergence without overfitting.

5. Batch Size

Tested 4, 8, and 16.
Batch size 8 provided balanced performance.

Impact of Tuning

After tuning:

- Loss reduced faster.
- Accuracy stabilized.
- The model generalized better on test data.